Aerodynamic Development of Front & Rear Wing

1. The motivation behind creating the front and rear wings for the 2026-2027 FSAE Racing Car lies in the remarkable performance benefits the aerodynamic package yields. Essentially, you can increase the loading in both front and rear tires by substantial margins by only adding a minor amount of weight to the overall vehicle. Not having said front and rear wings on an open-wheeled racecar doesn't address the drag the wheels cause, and creates no loading on the tires during cornering, which limits the vehicles' overall ability to perform. The sole performance goal for a race car is to have its fastest possible time around the track, and that goal is more achievable with a front and rear wing. In both professional open-wheel, single-seater racing championships, which are F1 and IndyCar, teams invest great amounts of resources into developing their front and rear wings. The increase in tire loading mentioned earlier improves both braking and cornering performance, as the lateral acceleration that can be achieved is much greater.

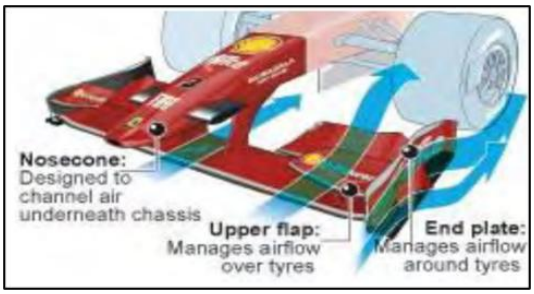


Figure 1. Front wing airflow

2. For the past 10 years cougar racing has been attending SAE International Competition without a full aerodynamic package. The absence of a front and rear wing affects vehicle dynamic performance as the vehicle can experience over and under steering. The aerodynamic wings will increase the vertical load on the front and rear tires. This increased load will provide a greater grip and lead to more stability at greater cornering speeds and an overall faster lap time. The goal is to produce 350 N of downforce at 55 mph, have a Center of Pressure of 55-60% distance of the front axle and maintain the weight of under 22 lbs. The project will require the investigation and prototyping of different manufacturing methods for composite materials. Realistic CFD simulations are also another thing to consider; the team needs to develop advanced 3D simulations and ensure their validity. A third consideration will be, can the design be validated properly using an accessible wind tunnel or any other method. A list of deliverables will determine the ideal efficiency (lift/drag), CFD vs experimental correlation, and finding a manufacturing route that is feasible. Ultimately, the question to solve this project is whether the aerodynamic package can provide an improvement that is measurable while meeting constraints such as weight, cost, and manufacturability. We need to be able to prove that the gains we can see outweigh the constraints

3. The main challenges the team have identified to be the most critical are the fabrication process and validation. We have knowledge that the fabrication section of the project will by far be the most difficult, and for good reason. We currently have our wings materials set to carbon fiber, which is a material that provides optimal strength to weight ratio and isn't as readily available and manufacturable as steel or traditional metals such as aluminum and brass. In addition, it has a far different composition and creation process, which is a big hurdle the team must overcome. However, since most aerodynamic elements on racecars are typically made of carbon fiber, due to teams wanting to increase vertical loading on the tires without adding on too much weight to the car, there are lots of valuable resources out there to help us on our journey. The next hurdle we are anticipating being extremely difficult is the validation process. Due to atmospheric air being invisible to the naked eye, physically seeing our wings do what they are supposed to do is impossible. Also, we are not easily available to see a clean downforce number while the car is racing around the track. Therefore, we will rely heavily on CFD simulations to give us numbers such as downforce, fluid velocities, and lift/drag coefficients. There are some also some methods we could use, those being Measurement of Lift: Suspension travel for each of the wheels Corresponding relative displacements of the suspension components can be translated into equivalent lifts and the fact that aerodynamic load on the suspension can also be measured by strain gauges on the springs. The key to overcoming these challenges to the best of the team's ability is to manufacture the components of both front and rear wing as close as possible to the physical properties used during the simulation trials.

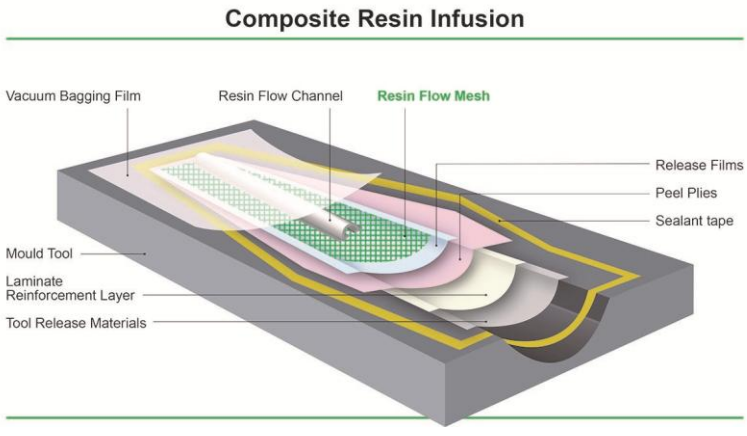


Figure 2. Manufacturing method of resin infusion

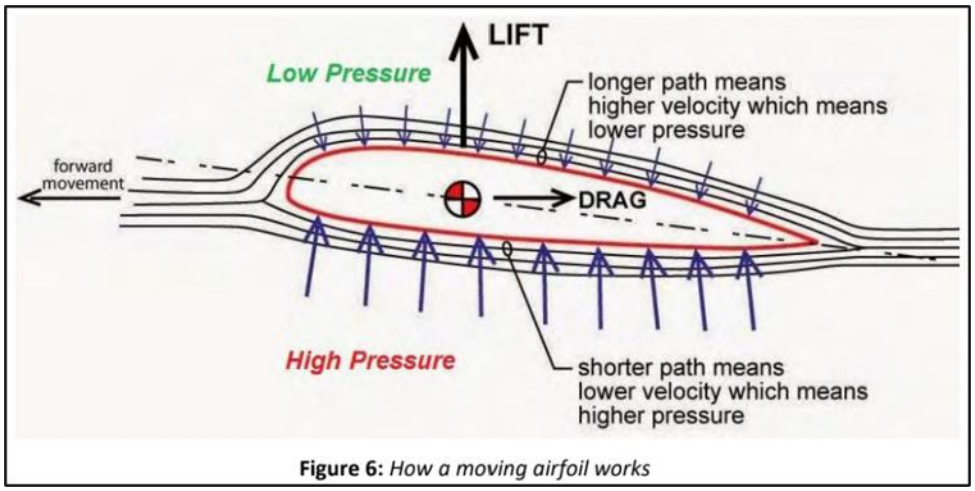


Figure 6: *How a moving airfoil works*

Figure 3. Aerodynamic principal of lift generation